

CELL CONSTRUCT A Zn-Cu ELECTROCHEMICAL CELL AND VALIDATE THE OUTPUT BY CONNECTING THE LED LIGHT.

Exp. No. :

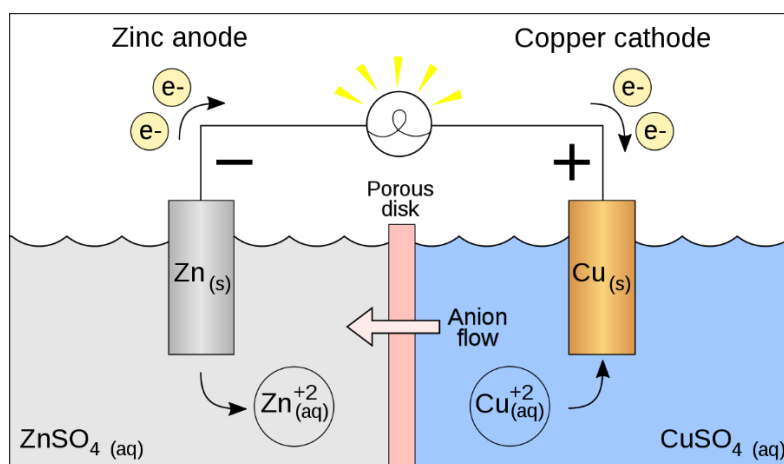
Date :

AIM/ OBJECTIVE:

To construct an electrochemical cell having zinc-copper and measuring its output voltage.

PRINCIPLE:

If we connect the zinc and copper by means of a metallic conductor, the excess electrons that remain when Zn^{2+} ions emerge from the zinc in the left cell would be able to flow through the external circuit and into the right electrode, where they could be delivered to the Cu^{2+} ions which become "discharged", that is, converted into Cu atoms at the surface of the copper electrode. The net reaction is the oxidation of zinc by copper(II) ions:



MATERIALS REQUIRED:

1M Zinc sulphate solution, copper sulphate solution, zinc rod, copper rod, beakers, voltmeter or potentiometer, salt bridge, connecting wires.

PROCEDURE:

- In a U-shaped tube heat 20gm of agar-agar gel and 5gm of potassium chloride in a beaker.
- Suck the molten mixtures into U-shaped tube.
- Allow it to cool.
- Prepare 1M of zinc sulfate solution and 1M of copper sulfate solution in separate beakers.
- Connect a salt bridge in such a manner that its one end dips in zinc sulfate and the other dips

in copper sulfate solution.

- Now connect the zinc plate to negative terminal of voltmeter and copper plate to positive terminal of voltmeter.
- Note down the reading of voltmeter in observation table.
- Take 10ml of 1M zinc sulfate and add water to make its volume up to 100ml. The resultant molarity will be 0.1M.
- Now dip zinc rod in 0.1M zinc sulfate solution and copper rod in 1M copper sulfate solution and note the readings.
- Repeat the same for 0.1M and 0.01M of zinc sulfate solution.
- Note down the reading in voltmeter.

RESULTS & DISCUSSION:

The output voltage of zinc copper cell is _____ V.

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
2	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	100	
Evaluator's Signature		